Cellular and Medical Immunology: Acute, Chronic Inflammation: Mediators and Modulators

Integrated Biomedical and Clinical Sciences OMFS 513

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CALIFORNIA NORTHSTATE UNIVERSITY COLLEGE of DENTAL MEDICINE

Mission: To advance the science and art of health through education, service, scholarship and social accountability

Learning Objectives

1. Describe the inflammatory process.
2. Compare and contrast acute and chronic inflammation.
3. Describe the tissue repair process.
4. Describe the mediators of inflammation and the therapeutic approaches that modulate the immune response.

Outline

Definition and etiology of inflammation

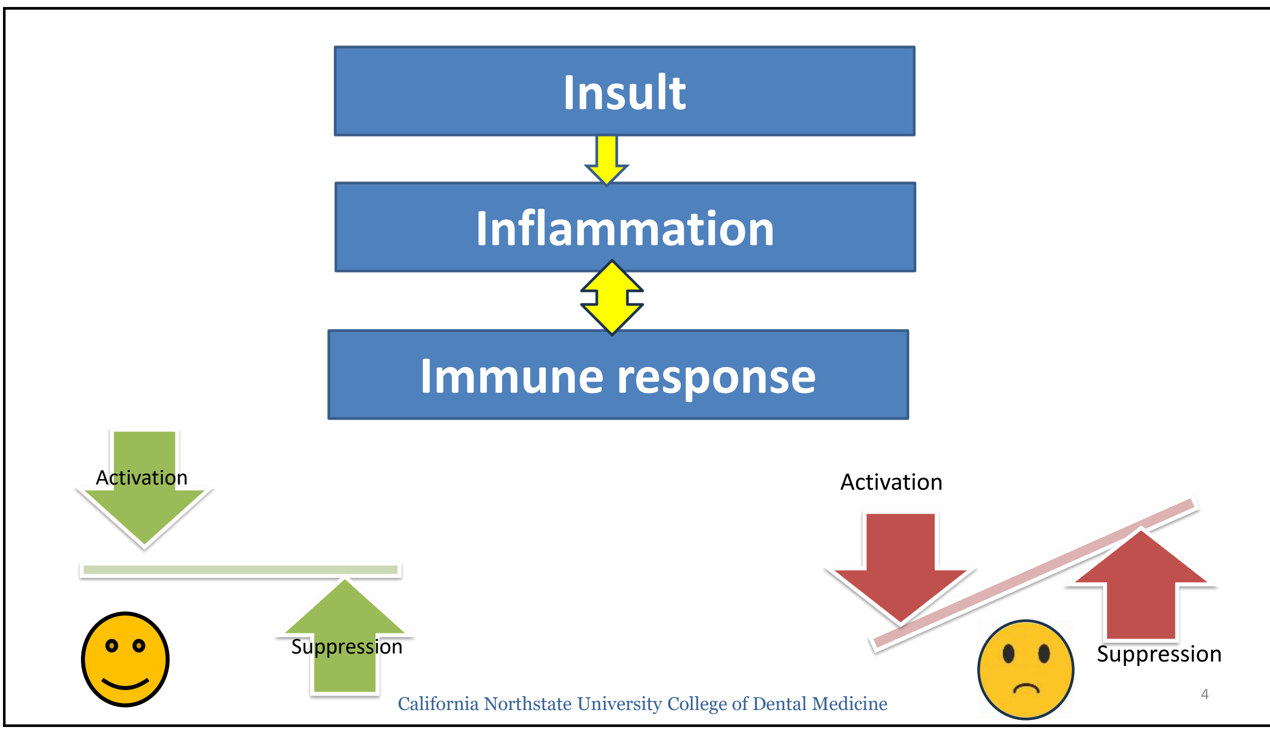
Acute inflammation

Chronic inflammation

Wound healing

Mediators of inflammation.

Modulators of inflammation.


Insult
Inflammation
Immune response
Activation
Suppression
Activation
Suppression
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Inflammatory Response

- Protective RAPID & NONSPECIFIC response of vascularized tissues to insults.
- Inflammation release molecules and brings immune cells from circulation to the inflammation site (chemical+ vascular+ cellular).
- Aims to eradicate the initial reason of cell injury, eliminate any offending agent and restore homeostasis for disrupted host tissues.
- Must be tightly regulated. Uncontrolled inflammation can cause chronic inflammatory conditions.

Inflammatory responses and inflammation-associated diseases in organs - PMC (nih.gov)

Causes of Inflammation

- **Microbial:** infections or microbial toxins.
- **Injury:** ischemic, chemical injury, or physical injury can lead to cell death & initiate an inflammatory response.
- **Foreign bodies:** splinters that breach the physical barriers. Inflammation due to tissue injury or introduction of microbes.
- **Immune reactions:** hypersensitivity or autoimmune reactions (immunologic injury).

Sequence of the inflammatory process

5 "R"

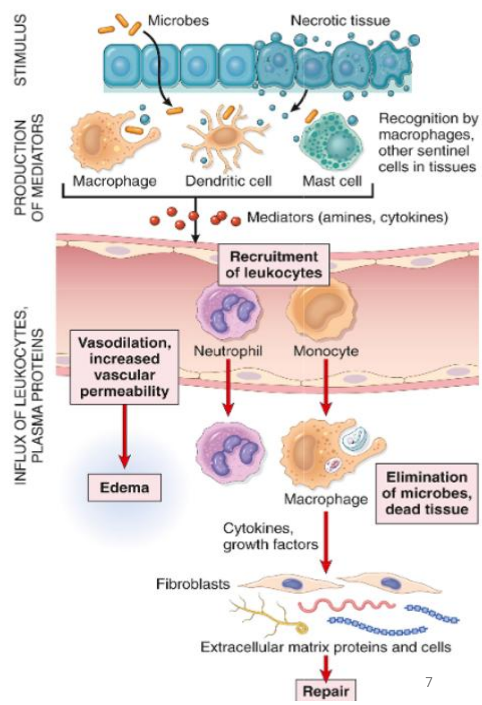
1- Recognition.

2- leukocyte Recruitment.

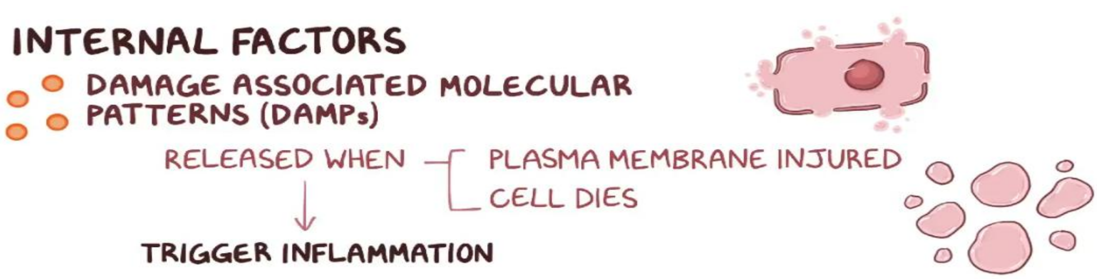
3- Removal of the threat (stimulus).

4- Regulation (Termination of inflammation).

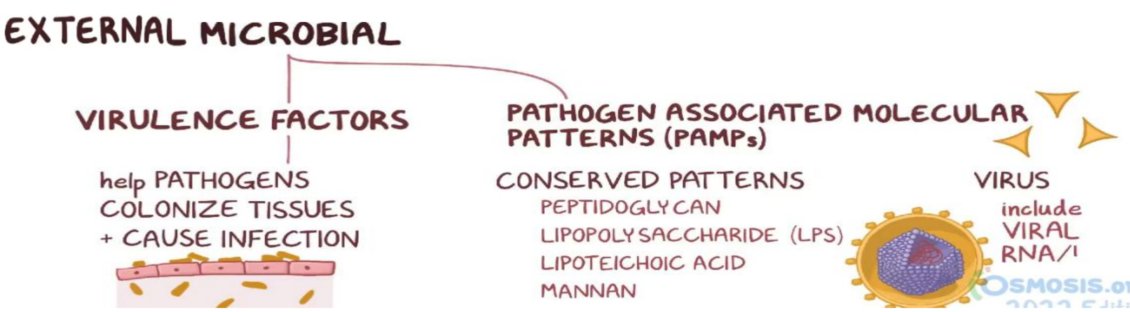
5- Tissue Repair: regain normal tissue or fibrous tissue/scar.



Recognition



NOD-like receptors: Sensors of cell damage detect DAMPs & activate the inflammasome.
Inflammasomes induces the production of interleukin-1 (IL-1) and IL-18.
IL-1 recruits leukocytes and induces inflammation.

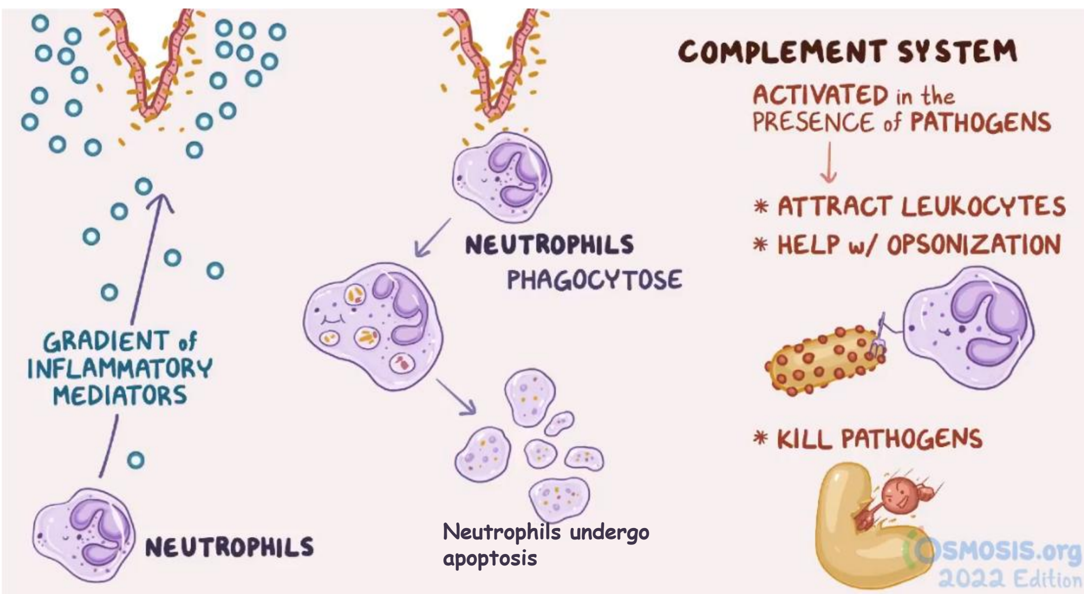


- Circulating complement proteins and TLR
- Other proteins include "collectins" that bind to microbes and promote their phagocytosis.

Leukocyte recruitment
via inflammatory mediators

- Tissue resident cells release inflammatory mediators that recruit other leukocytes.
- During the first 6 to 24 hours, neutrophils are main component of the inflammatory infiltrate. After 24h to 48 hours, neutrophils are gradually replaced by macrophages.
- Neutrophils, monocyte-derived macrophages and complement proteins are the main players in most inflammatory responses.
  - **Histamine and bradykinin** increase vascular permeability.
  - **IL-6 and TNF-α** stimulate the production of C-reactive protein from the hepatocytes. In turn, the CRP activate the production of complement proteins.

Removal of Threat



Regulation: Termination of Inflammation

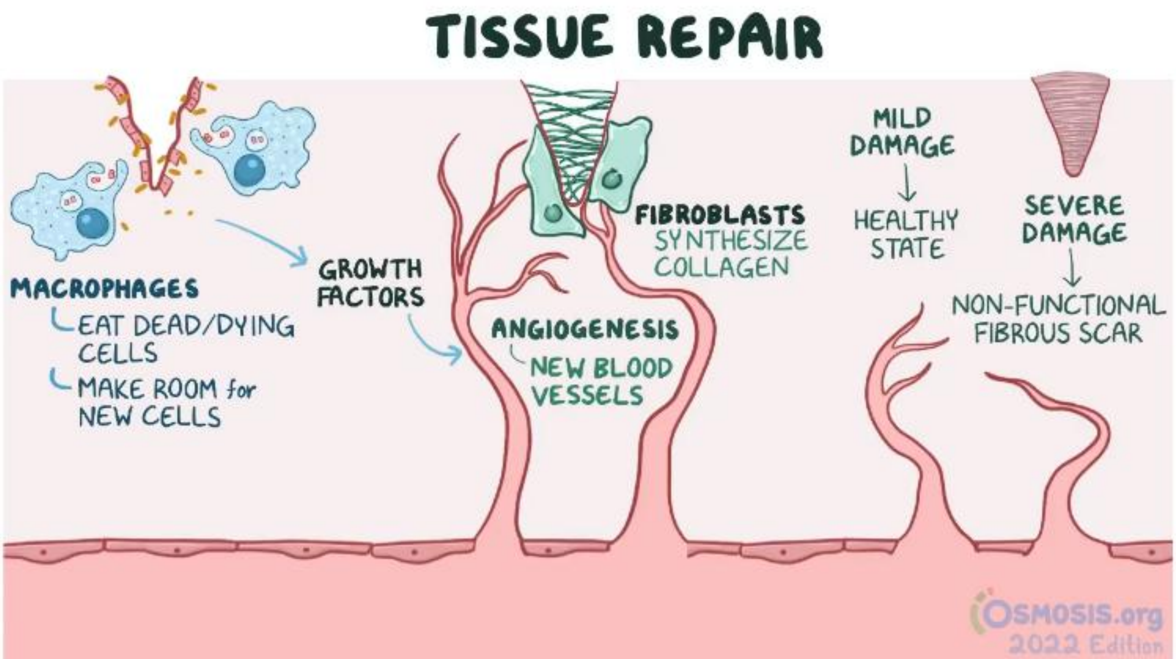
Inflammation should be terminated in timely fashion to prevent excessive tissue injury

- Inflammation declines after the offending agents are removed.
- Several termination mechanisms like

1- Switching the arachidonic acid pathway from proinflammatory leukotrienes production to anti-inflammatory mediators (lipoxins).

2- Release of anti-inflammatory cytokines like transforming growth factor-β (TGF-β) and IL-10 from macrophages and other immune cells.

3- Neural impulses can also inhibit TNF production from macrophages.



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Inflammation

- Acute Inflammation
- Chronic Inflammation

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Acute Inflammation

Initial and rapid response to infections and tissue damage.

Hallmark of acute inflammation: vascular events and cellular responses which are activated by inflammatory mediators derived from various leukocytes.

Acute inflammation

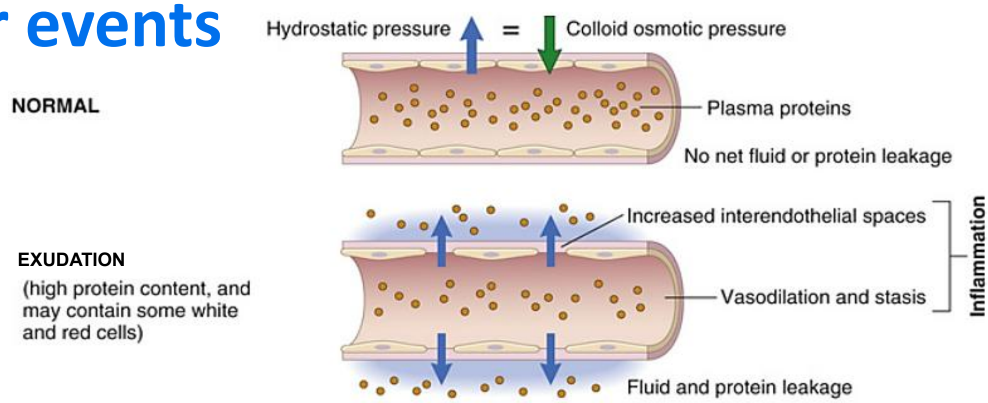
(1) Dilation of small vessels, leading to an increase in blood flow.

(2) Increased permeability of the microvasculature to enable plasma proteins and leukocytes exit the circulation to the injury site.

(3) Emigration of leukocytes from the microcirculation, accumulation at the site of inflammation, activation and elimination of the offending agent.

Platelet-activating factor, produced by leukocytes, activated mast cells. Cause vasodilation, increased permeability, leukocyte adhesion, chemotaxis, platelet activation

Vascular events

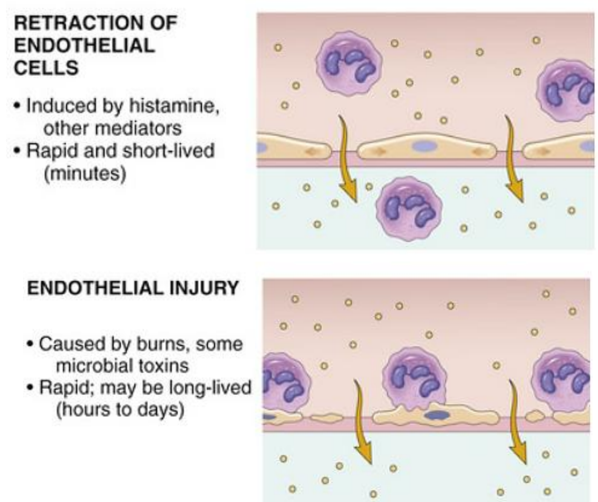


Vascular events may include transient vasoconstriction (secondary to trauma) followed by vasodilation, increased vascular permeability and exudation. Leads to increased blood flow ➡ flow of plasma proteins and leukocytes to the injury site.
Exudate has high protein content and contain cells/cell debris (inflammation related)
Transudate has low protein content and little or no cells (Liver disease, CHF related).

Mechanism of Increased Vascular Permeability

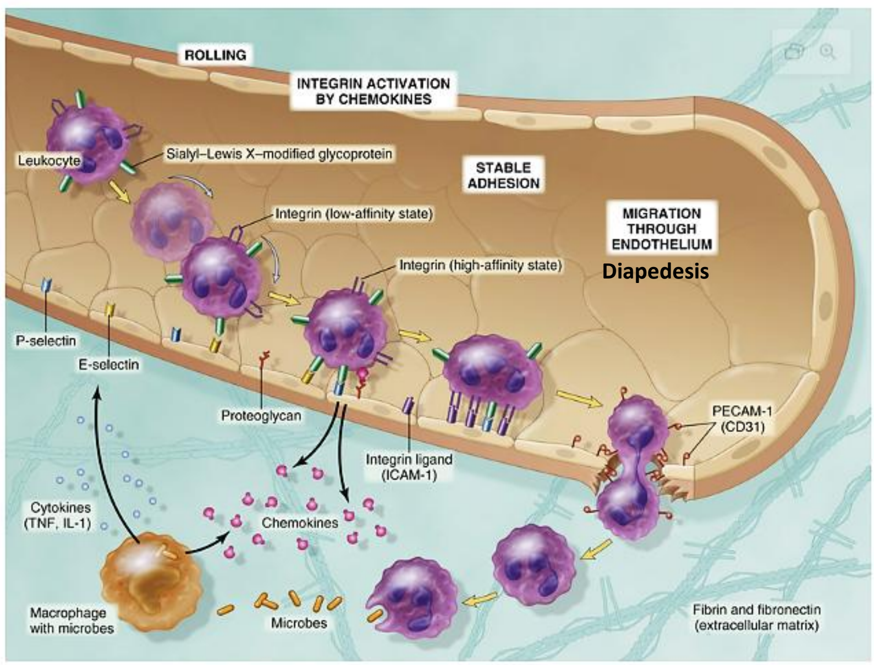
1- Retraction of endothelial cells (most common): “aka, immediate transient response”. Secondary to inflammatory mediators. Rapid onset and short duration.

2- Endothelial injury: secondary to direct damage to the endothelial cell layer. Exudation starts immediately after injury and is sustained until the damaged vessels are repaired.



Cellular Events

- Emigration (margination, rolling, adhesion & transmigration).
- Chemotaxis.



Selectins mediate initial weak interactions between leukocytes and endothelium. **Integrins-ICAM** mediate firm adhesion of leukocytes to endothelium.

ICAM= Intercellular adhesion molecule
PECAM= Platelet endothelial cell adhesion molecule

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1- Emigration: it refers to the passage of the inflammatory leukocytes between the endothelial cells into the adjacent interstitial tissue. The process of emigration involves the following multiple steps:

a. Margination which denotes the localization of the leukocytes to the blood vessel periphery. By moving close to the vessel wall, leukocytes can detect changes in the endothelium.

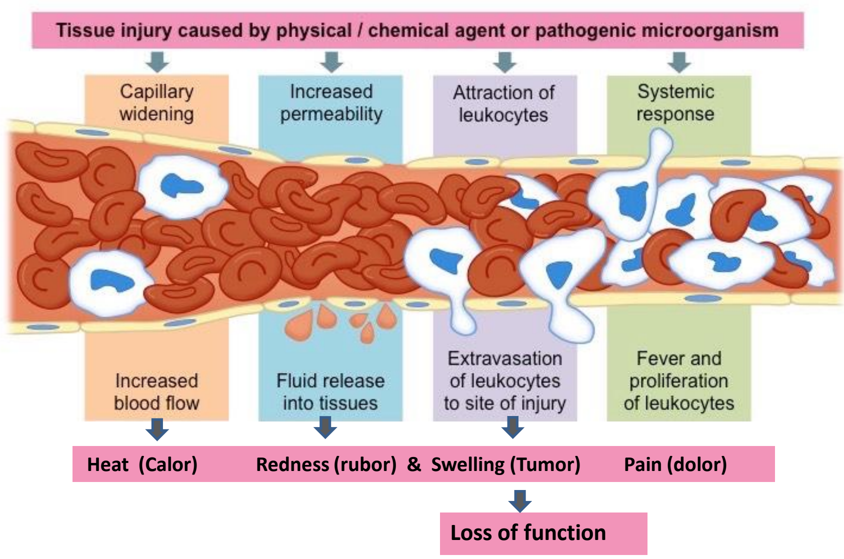
b. Rolling (or tumbling), this refers to the loose attachment of the leukocytes to the blood vessel wall. Rolling is mediated by the action of endothelial selectins.

c. Adhesion occurs as leukocytes adhere to the endothelial surface and it is mediated by the interaction of integrins on leukocytes with the endothelial ligands like "ICAM".

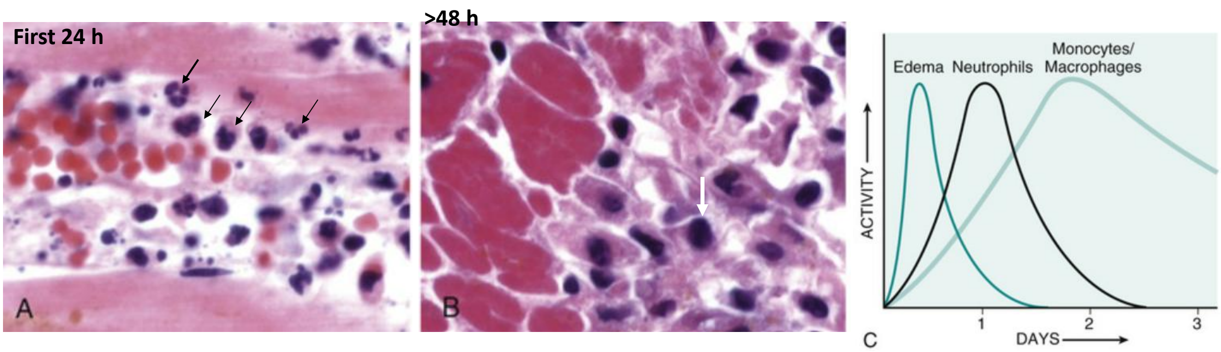
d. Transmigration (also termed **diapedesis**) is the movement of leukocytes across the endothelium.

2- **Chemotaxis** is mediated by chemotactic chemical compounds that attract different leukocytes.

Cardinal symptoms of Inflammation



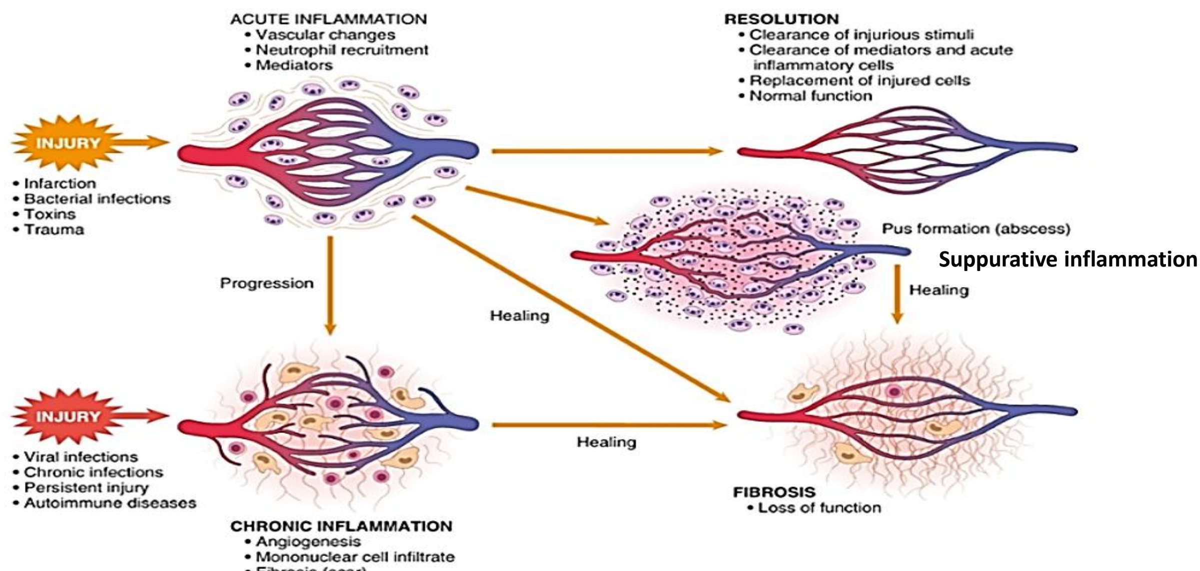
Nature of leukocyte infiltrates



Physiological Benefits

1. Prevent infection and tissue damage caused by invading microorganisms.
2. Prepare for healing and repair after the eradiation of microbes or dead cells/debris.
3. Initial innate defense facilitates the development of adaptive immunity.

Outcome of Acute Inflammation



Inflammation

Acute Inflammation

Chronic Inflammation

Chronic inflammation

- Prolonged duration (weeks or months).

May follow acute inflammation or may begin without preceding acute reaction. Chronic inflammation contributes to the pathogenesis of different disorders.

Causes

- Persistent and recurrent infection or prolonged exposure to toxins.
- Hypersensitivity diseases.
- If the initial inflammatory reaction was insufficient to neutralize the insult.
- *De novo*, without a preceding acute inflammatory reaction (idiopathic) .

Morphologic Changes

Chronic inflammation is characterized by:

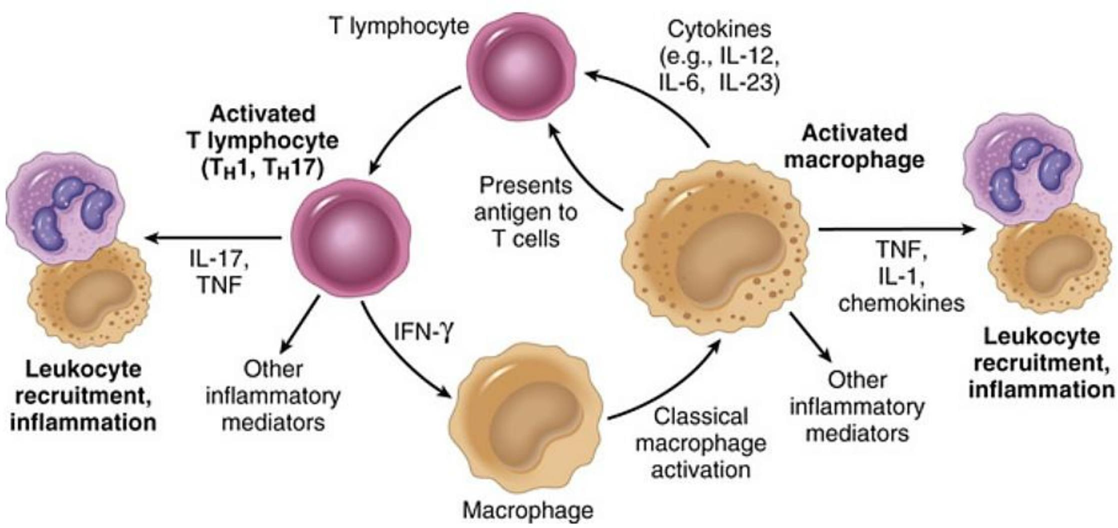
a) Infiltration with mononuclear leukocytes (macrophages, lymphocytes and plasma cells) leading to continuous inflammation.

b) Tissue injury induced by the persistent inflammation.

c) Attempted repair. This is achieved the replacement of the damaged tissue with connective tissue, angiogenesis and scarring.

In most chronic inflammatory reactions, macrophages and lymphocytes are the dominant population. They secrete cytokines and growth factors and activate other cells.

Cellular Mediators



Patterns of chronic inflammation

Chronic nonspecific inflammation

Mediated by the interaction of monocytes–macrophages with lymphocytes. Persistent insult stimulate the activation of B and T-cells. In turn, the lymphocytes maintain and propagate the inflammatory process. Proliferation of fibroblasts, connective tissue, and epithelial cells is observed leading to scar formation.

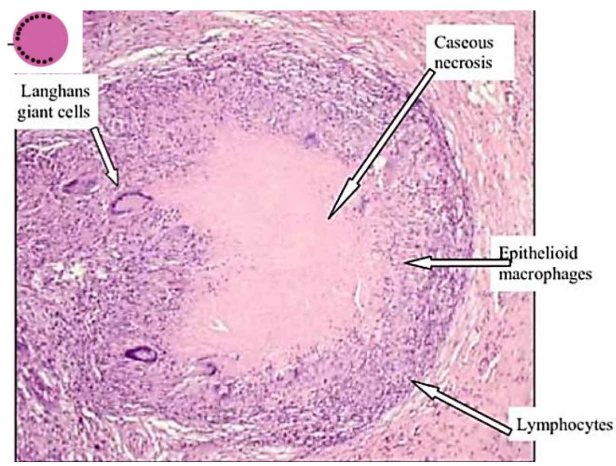
Granulomatous inflammation

Induced by T cell and macrophage activation in response to an insult resistant to eradicate.

Aim: to contain the insult/pathogen.

Histology: Activated macrophages develop abundant cytoplasm, named epithelioid cells. Some activated macrophages fuse to form multinucleated cells (Langhans giant cells). Caseating and non-caseating granulomas.

Example: bacterial Infections (like tuberculosis), fungal (like coccidioidomycosis), autoimmune/ inflammatory conditions (like Crohn disease)



Acute versus Chronic Inflammation

Feature	Acute	Chronic
Onset	Fast: minutes or hours	Slow: days
Cellular infiltrate	Mainly neutrophils	Monocytes/macrophages and lymphocytes
Tissue injury, fibrosis	Usually mild and self-limited	May be severe and progressive
Local and systemic signs	Prominent	Less

Systemic effects and Diagnostic markers

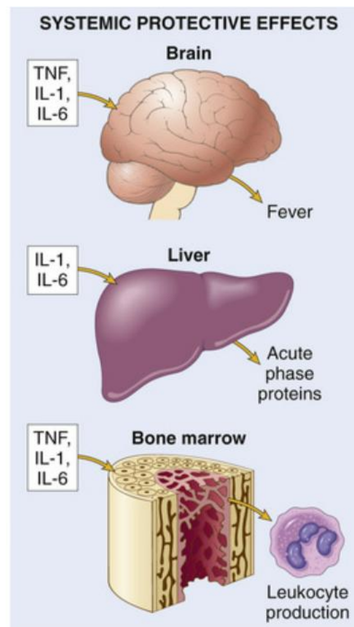
Systemic effect of inflammation

TNF, IL-1, and IL-6 are responsible for the clinical manifestations of inflammation

1. Fever

TNF and IL-1 act as endogenous pyrogens (IL-6, less potent pyrogen).

Trigger the synthesis and release of prostaglandins (PGE2) in hypothalamic cells. Consequently, this will raise the body's steady-state temperature to a higher level though inducing vasoconstriction (decrease heat loss) and affect skeletal muscle and fat metabolic rate (increase heat generation).

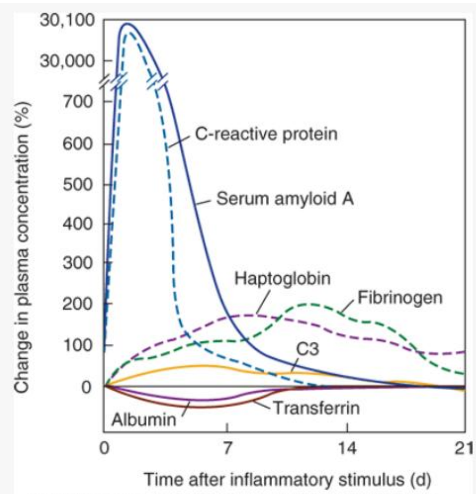


2. Acute phase proteins

Elevated levels can serve as diagnostic markers for inflammatory diseases.

Inflammatory mediators (especially, IL-6) induce the release of acute-phase proteins from the liver. C-reactive protein, fibrinogen, and serum amyloid A are commonly elevated in inflammatory reactions.

CRP and SAP promote and modulate the immune response in inflammation. Fibrinogen may contribute to hemostasis, tissue repair...



Source: K.E. Barrett, S.M. Barman, H.L. Brooks, Jason X.J. Yuan: Ganong's Review of Medical Physiology, Twenty-Sixth Edition

3. Leukocytosis

Increase in the leukocyte count in response to inflammatory reactions (normal WBCs<11,000/ul; Infection WBCs>15,000/ul). TNF-α, IL-1, IL-6 and other colony stimulating factors stimulate leukocyte production and accelerate their release from the bone marrow. Consequence? ----

4. Sepsis

Severe bacterial infections can stimulate massive cytokine production leading to the development of septic shock. Characterized by hypotensive crisis, fast respiratory rate, metabolic disturbance and coagulation disorders.

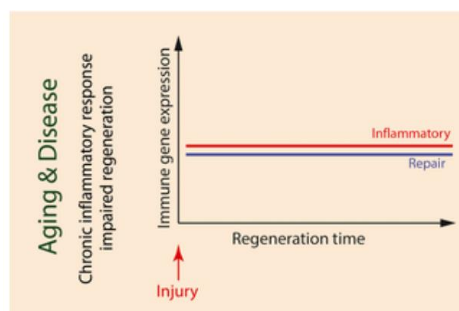
Diagnostic markers

- **Erythrocyte sedimentation rate:** Rate at which erythrocytes suspended in plasma fall when placed in a vertical tube (mm/hour). Indirect measure of the acute phase response. Normal levels 1–13 mm/hr for males and 1–20 mm/hr for females.
- **C-reactive protein:** Acute phase proteins made by the liver. The serum concentration in most people is under 3 mg/L. Increases in most inflammatory conditions. Highly elevated CRP levels are strongly associated with infection. CRP and ESR are elevated as part of the acute phase response.
- **CBC with differential**
- Elevated level of **inflammatory cytokines** (like, TNF α and IL-6).

Immunosenescence and Inflammaging

- Immunosenescence: changes that occur in the immune system with increasing age.
- Inflammaging: state of chronic, sterile, low-grade inflammation that develop during aging. Constant low-level production of circulating inflammatory mediators.

Immunosenescence and inflammaging are linked to several pathological conditions like Parkinson's disease, cardiovascular diseases, cancer and autoimmune disease.



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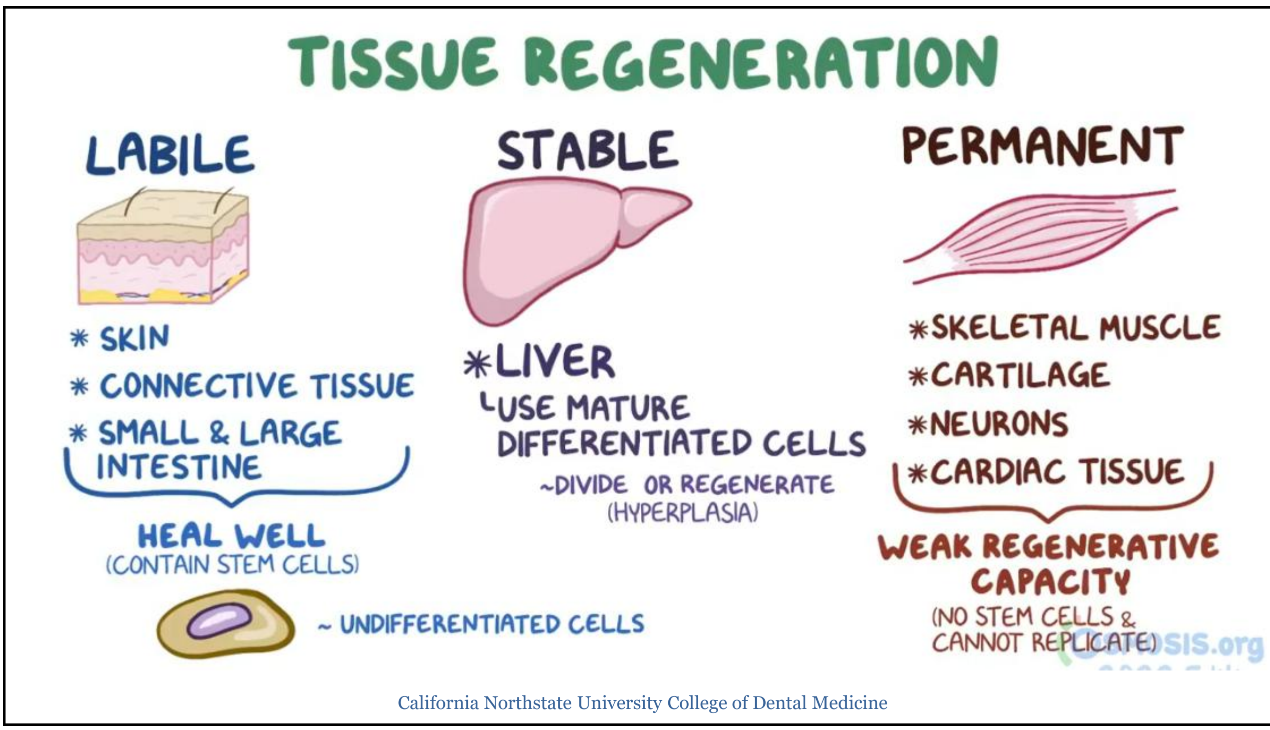
Tissue repair
Wound healing


TISSUE REGENERATION
LABILE
* SKIN
* CONNECTIVE TISSUE
* SMALL & LARGE INTESTINE
HEAL WELL
(CONTAIN STEM CELLS)
~ UNDIFFERENTIATED CELLS
STABLE
*LIVER
└USE MATURE DIFFERENTIATED CELLS
~DIVIDE OR REGENERATE
(HYPERPLASIA)
PERMANENT
*SKELETAL MUSCLE
*CARTILAGE
*NEURONS
*CARDIAC TISSUE
WEAK REGENERATIVE CAPACITY
(NO STEM CELLS & CANNOT REPLICATE)
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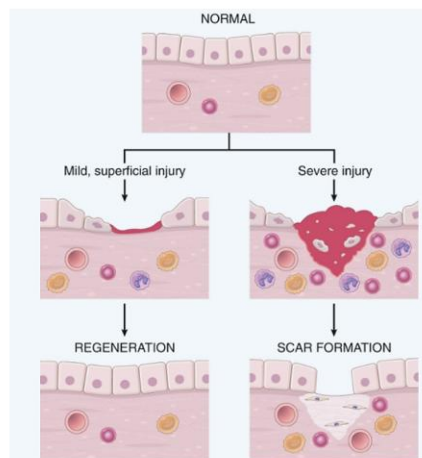
Modes of tissue repair

- Regeneration

Involves complete replacement of the damaged cells without scar formation.

- Connective tissue deposition (scar formation)

Involves the regeneration of cells combined with scar formation and fibrosis.



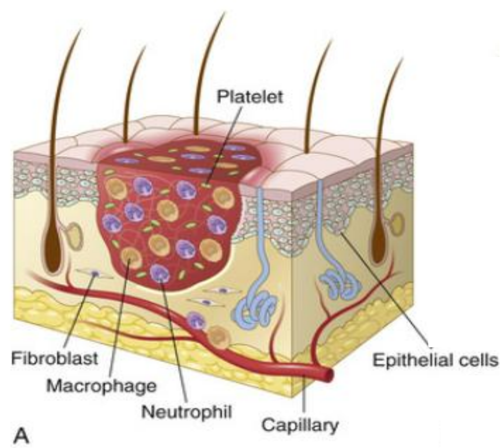
Steps of tissue repair "Scar formation"

Hemostasis and Initiation of Inflammation Phase

i) Platelets form platelet plug to initiate hemostasis.

ii) Inflammation recruit neutrophils and then macrophages. Macrophages play multiple roles:

  a. M1 macrophages clear microbes and necrotic tissue and promote inflammation.
  b. M2 macrophages produce growth factors that stimulate the proliferation of cell types involved in the repair process.

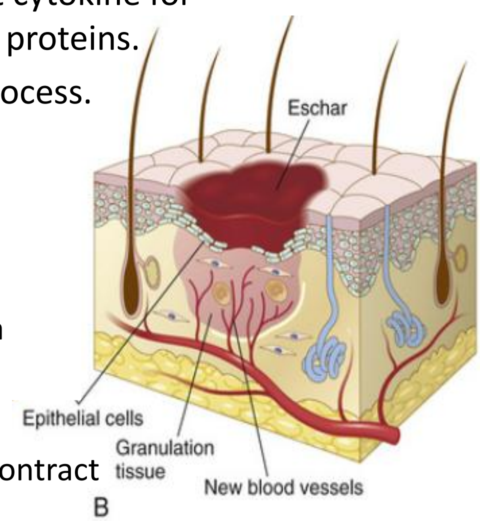


Proliferation phase: Formation of granulation tissue.

Different cell populations work together to ensure healing and form the granulation tissue. **TGF-β** is an important cytokine for the synthesis and deposition of connective tissue proteins.

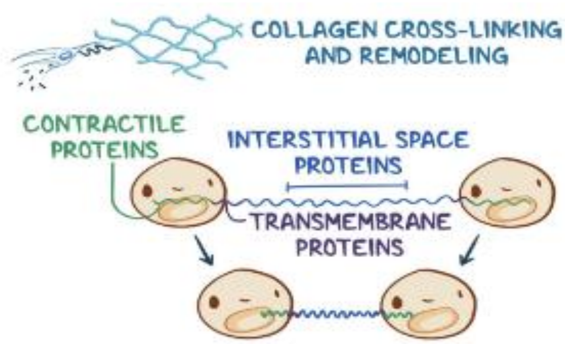
Granulation tissue is unique for wound healing process. Composed of:

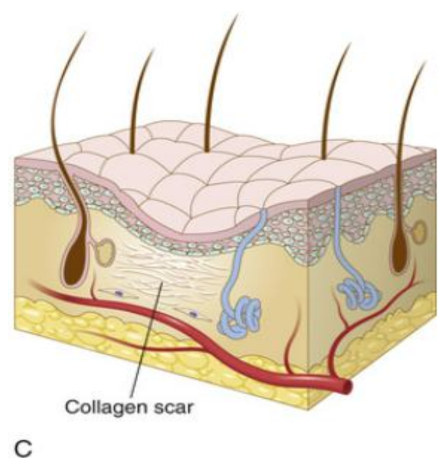
- Proliferating fibroblasts secrete collagen (III), glycoproteins to make ECM.
- Loose connective tissue.
- Endothelial cells that form new blood vessels via angiogenesis.
- Scattered chronic inflammatory cells.
- myofibroblasts produce contractile proteins to contract the wound edges.

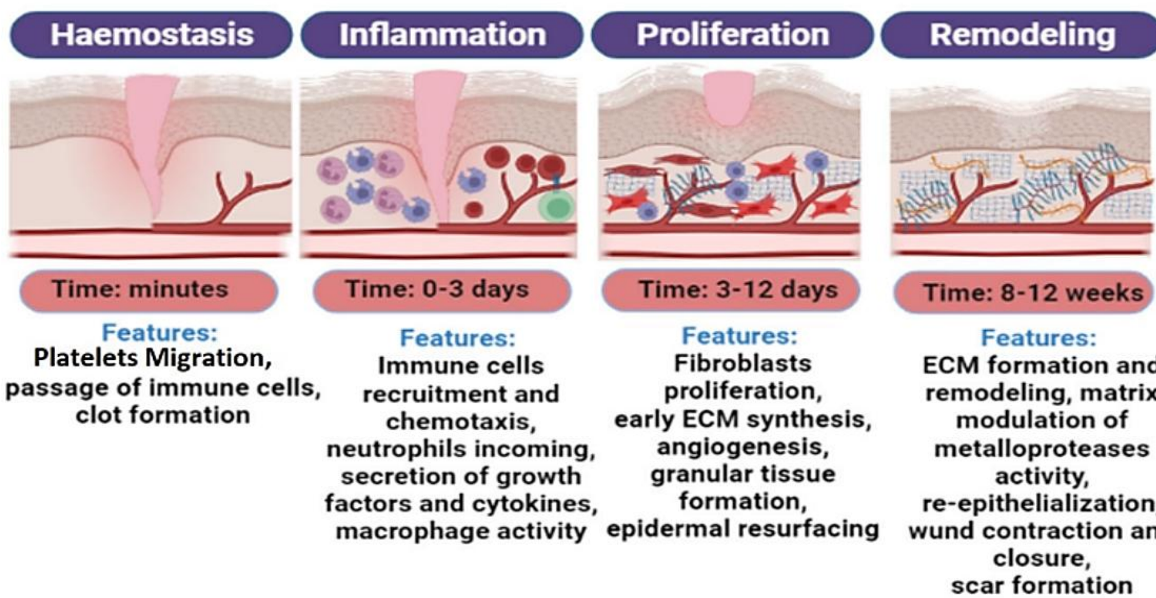


Remodeling Phase

In this step, fibroblasts proliferate and continue depositing collagen fibers that crosslinks to increase the tensile strength to produce a stable fibrous scar.







(PDF) In Vitro and In Vivo Characterization Methods for Evaluation of Modern Wound Dressings California Northstate University College of Dental Medicine

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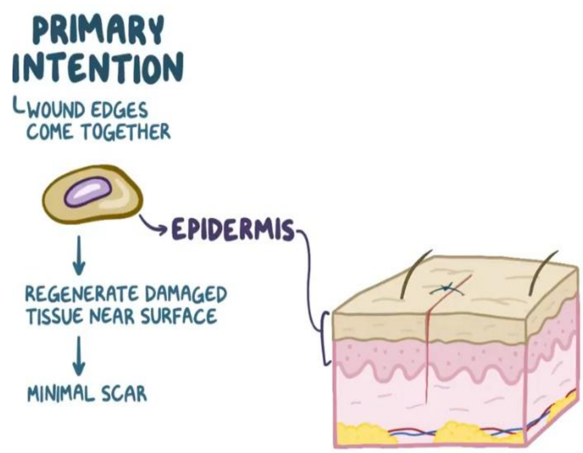
Closure of the Wounds

Healing by Primary (First) Intention

Example: Healing of most surgical incision. Wound has clean edges.

Mechanism: Epidermis or uppermost tissue layer can regenerate the damaged tissue.

Result: Small to non-existent scar.

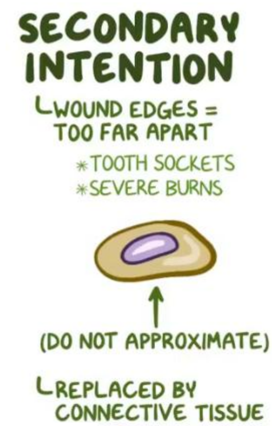


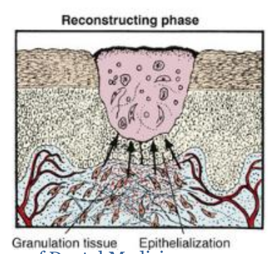
Healing by Second (secondary) Intention

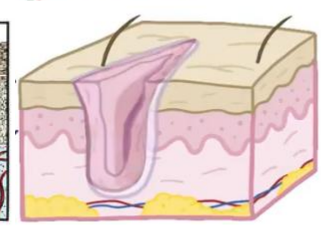
Example: Healing of a larger wounds with unclean edges or tooth extraction.

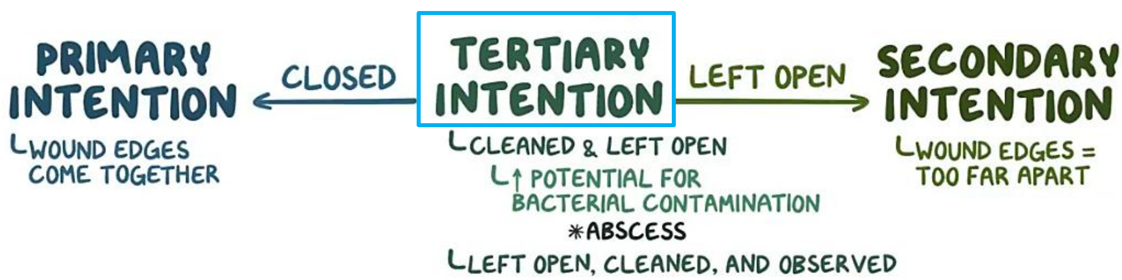
Mechanism: Wound edges are far apart. The wound is replaced by connective tissue that grows from the base of the wound upwards.

Result: Larger, more prominent scar









Tertiary intention (delayed closure), refers to wounds that were cleaned and left open to be monitored.

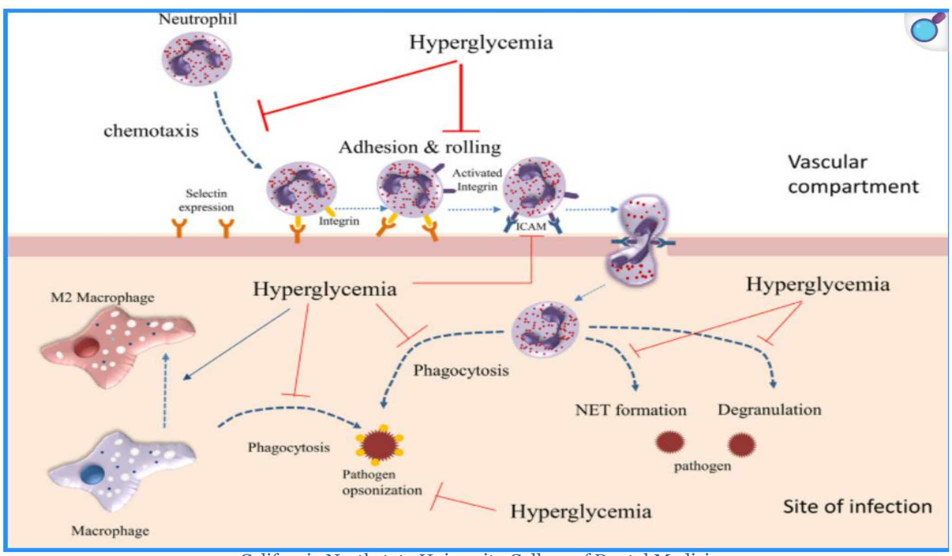
- If closed with clean margin → healing by first intention.
- If left open → healing by secondary intention

Factors that impede repair

The depth and location of injury determines the time required for repair.

- Persistent Infection
- Diabetes.
- Dietary deficiency (like protein or vitamin C)
- Glucocorticoids (partially via inhibiting TGF-β).
- Others: aging, smoking, poor blood supply and stress.

Impairment of Immune response with Hyperglycemia



Mediators of Inflammation

Mediators of Inflammation

Can initiate and regulate the inflammatory reaction. Can serve as therapeutic targets to modulate the inflammatory response.

Inflammatory mediators are either cell-derived or plasma derived.

- Vasoactive amines like Histamine.
- Eicosanoids (Prostaglandins and leukotrienes).
- Cytokines and Chemokines
- Platelet-activating factor
- Complement and kinins.

Mediator	Source	Action
Histamine	Mast cells, basophils, platelets	Vasodilation, increased vascular permeability, endothelial activation
Prostaglandins	Mast cells, leukocytes	Vasodilation, pain, fever
Leukotrienes	Mast cells, leukocytes	Increased vascular permeability, chemotaxis, leukocyte adhesion, and activation
Cytokines (TNF, IL-1, IL-6)	Macrophages, endothelial cells, mast cells	Local: endothelial activation (expression of adhesion molecules). Systemic: fever, metabolic abnormalities, hypotension (shock)
Chemokines	Leukocytes, activated macrophages	Chemotaxis, leukocyte activation
Platelet-activating factor	Leukocytes, mast cells	Vasodilation, increased vascular permeability, leukocyte adhesion, chemotaxis, degranulation, oxidative burst
Complement	Plasma (produced in liver)	Leukocyte chemotaxis and activation, direct target killing (membrane attack complex), vasodilation (mast cell stimulation)
Kinins	Plasma (produced in liver)	Increased vascular permeability, smooth muscle contraction, vasodilation, pain

Vasoactive Amines

Histamine & Serotonin

- Histamine is stored in mast cells granules and released by degranulation (physical injury or IgE crosslinking).
- Histamine is also present in basophils, eosinophils (minor) and platelets.
- Histamine cause vasodilation and increase vascular permeability. Histamine activates the H_1 receptors on vascular endothelial cells.
- Anti-histamine drugs elicit their pharmacological response by blocking the H_1 receptor (antagonists).

Eicosanoids: Arachidonic acid metabolites

- Multiple biologic processes, including inflammation and hemostasis
- Phospholipase A2 stimulates the release of arachidonic acid.
- Arachidonic acid yield prostaglandins and leukotrienes via ----&----- pathway.
- Lipoxins, lipoxygenase pathway, suppress inflammation by inhibiting the recruitment of leukocytes.

Action	Eicosanoid
Vasodilation	Prostaglandins PGI_2 (prostacyclin), PGE_1, PGE_2, PGD_2
Increased vascular permeability	Leukotrienes C_4, D_4, E_4
Chemotaxis, leukocyte adhesion	Leukotriene B_4

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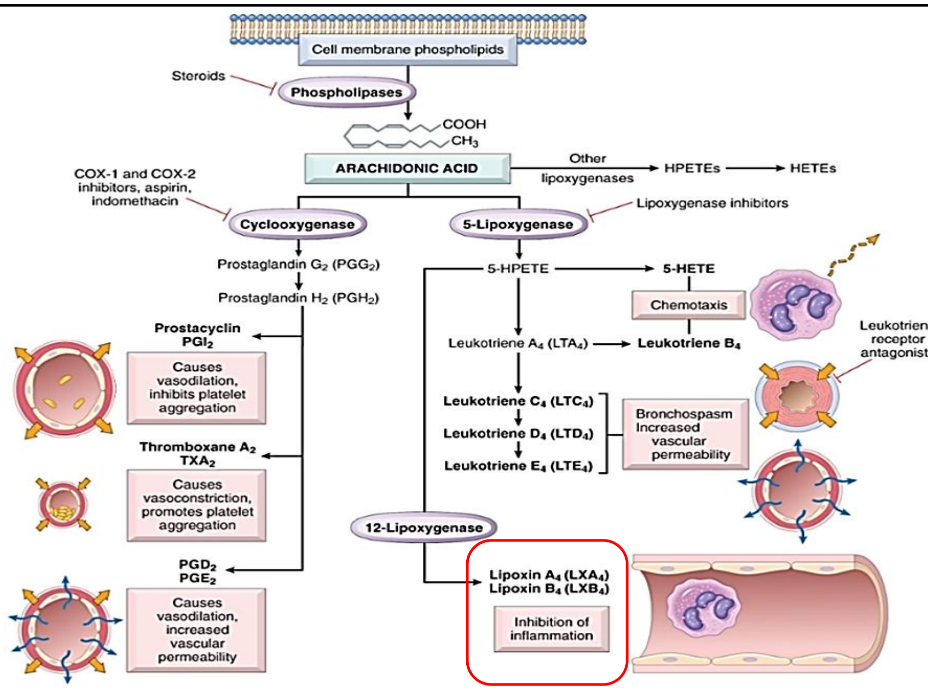

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Platelet-Activating Factor

For platelet aggregation and mediating inflammatory effects. Produced by several cells including platelets, basophils, mast cells, neutrophils and macrophages.

Kinins

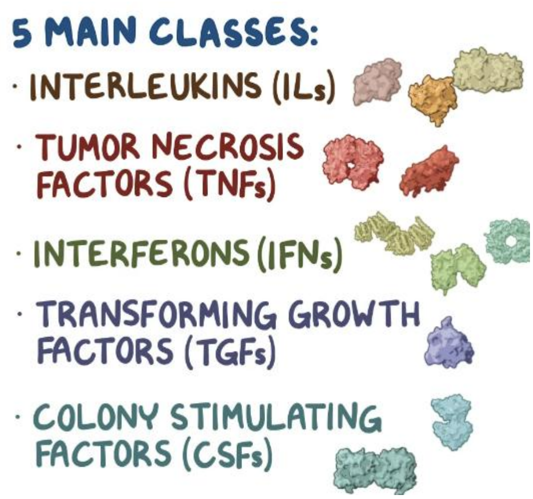
Bradykinin is derived from plasma protein precursor kininogens. Bradykinin increases vascular permeability and causes vasodilation and pain.

Chemokines

- Small sized-proteins that function as chemo-attractants to recruit leukocytes to the infection site.
- Secreted by macrophages, mast cells, T-cells, endothelial cells and others.
- Chemokines are important signaling molecules for cell migration or the recruitment of leukocytes to the site of inflammation (example neutrophile chemotactic factor, aka CXCL8 or IL-8, along with complement 5a and LTB4).

Cytokines

- Soluble mediators that initiate, terminate, and modulate the intensity of immune or inflammatory response.
- Small peptides that mediate their biological response through modulating cell surface receptors.
- Overproduction of (proinflammatory) cytokines can trigger the cytokine storm syndrome.



Tumor Necrosis Factors (TNFs)

Big family of pro-inflammatory cytokines with multiple roles.
Include: TNF-α and lymphotoxin-alpha (LT-α, aka TNF-β).
TNF-α:

- Produced by **macrophage**, T cells, mast cells, DCs and others.
- Can initiate inflammatory response and leukocyte recruitment.
- Increase vascular permeability.
- Induce fever
- Mediate neutrophil and macrophage induced-cytotoxicity of cancer cells.
- TNF can promote lipid and protein catabolism and suppress the appetite. Sustained production of TNF contributes to cachexia.
- Part of systemic acute-phase response

Interferons

- Type I:
  - Produced all **nucleated cells with viral infection** and **DCs, macrophages**.
  - Large family: **IFN-α, IFN-β,** IFN-ε, IFN-κ, and IFN-ω
  - IFN-α can stimulate innate responses, transcriptions of interferon stimulated genes (ISGs), activate NK cells to combat viral infection
- Type II (**IFN γ)**
  - Produced by immune cells (**NK and activated T-helper cell**). Activated by IL-12.
  - Functions: activate macrophages in chronic inflammation, anti-viral activity, multiple roles in innate and adaptive response.

COLONY STIMULATING FACTORS

GRANULOCYTE-MACROPHAGE COLONY-STIMULATING FACTOR (GM-CSF)

M-CSF for macrophages
G-CSF for neutrophils

HEMATOPOIETIC STEM CELLS

TRANSFORMING GROWTH FACTORS

- >30 DIFFERENT TGFs

T-regs
Promote wound healing

TGF-β

- INHIBITORY FACTOR
- Imp for wound healing
- SUPPRESSES PROLIFERATION & DIFFERENTIATION of T-cells, B-cells and macrophages (with IL-10)
- HELPS CD4+ T-CELLS become REGULATORY CELLS

Interleukins

Small proteins that activate and regulate the immune cell functions.

- **IL-1:** produced by most immune cells. It induces inflammation, fever "endogenous pyrogen" and activate TNF-α.
- **IL-2,** produced by T-cell and macrophages. Induce T cells proliferation and activation.
- **IL-4** induces the differentiation of Th2 lymphocytes & generate Th2 cytokines.
- **IL-6:** produced by macrophages, other immune cells. Induces B-cell stimulation, pro-inflammatory cytokine that stimulate the liver to secrete acute phase proteins.
- **IL-8:** by macrophages. Chemotaxis for different leukocytes (neutrophils).

- **IL-10:** T-cells & macrophages. **Inhibit** production of inflammatory cytokines.
- **IL-1, IL-6, and TNF-α** induce the acute phase response and are endogenous pyrogens.
- **IL-4, IL-5 and IL-13** mediate the inflammatory response to parasitic infections & allergic reactions.
- **IL-12** by activated macrophages and DCs cells, induce Th1 development. Th 1 generate more cytokines IL-2, interferon-γ, and TNFα to activate immune response.
- **IL-17**, by Th17. Pro-inflammatory

Modulators of Inflammation

Management strategies

1- Inhibit immune function/immunosuppression (corticosteroids).

2- Inhibit prostaglandin synthesis (cyclooxygenase pathway e.g: NSAIDs).

3- Histamine antagonists to block the histamine action.

4- Cytokine and cytokine receptor inhibitors (newer strategy, for example, Tumor necrosis factor inhibitor "Etanercept", FDA approved in 1998 for RA.

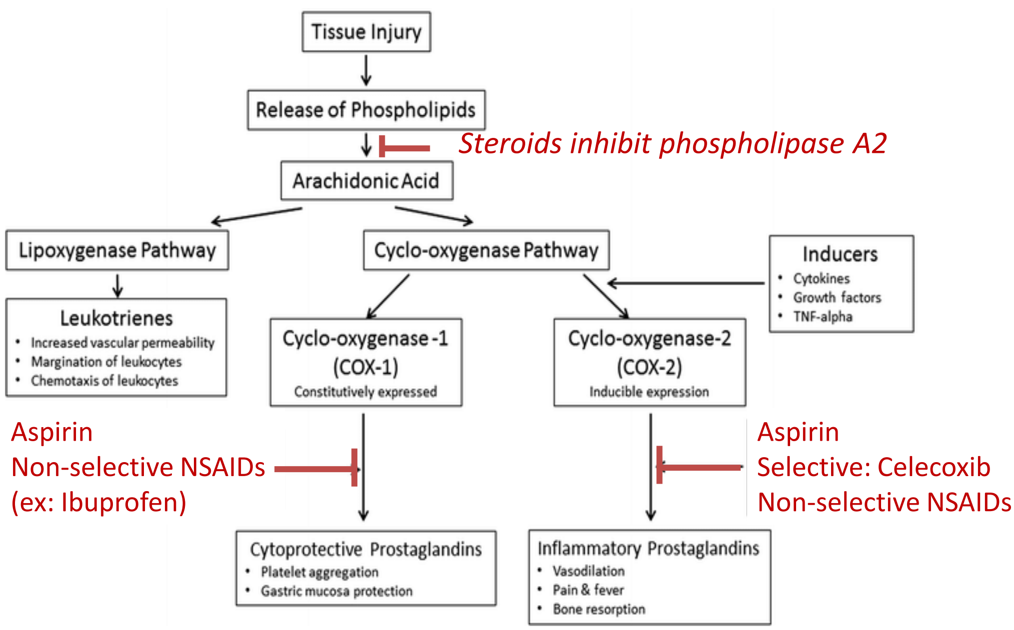
5- Local measures like using cold/ice packs to reduce swelling, pain and inflammation.

6- Platelet rich plasma

Platelet-rich plasma & Platelet-rich Fibrin (PRP vs PRF)

- Both are autologous blood-derived products
- **PRP**: Plasma that contain supraphysiological concentration of platelets and growth factors. Rapid release of growth factors.
- **PRF**: Blood spun at low speed with no anticoagulant, yielding fibrin gel clot that traps platelets and growth factors. Slower-steady release of growth factors.
- Both have growth factors, cytokines, and chemokines that modulate inflammation and promote faster healing with wide applications.
- Dental application: placement of these biomaterial in extraction socket can decrease alveolar bone resorption, increase bone density (implant success) and enhance healing

Autologous platelet concentrates in alveolar ridge preservation: A systematic review with meta-analyses - PubMed

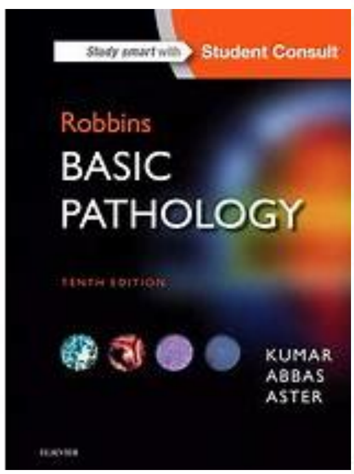


[Omega-3 fatty acids as an adjunct for periodontal therapy-a review - PubMed (nih.gov)]

Big picture

- Cardinal signs of inflammation. Vascular vs cellular events
- Different stages of inflammation and mechanisms of wound repair.
- Compare different aspects of acute and chronic inflammation.
- Mediators of inflammation

Resources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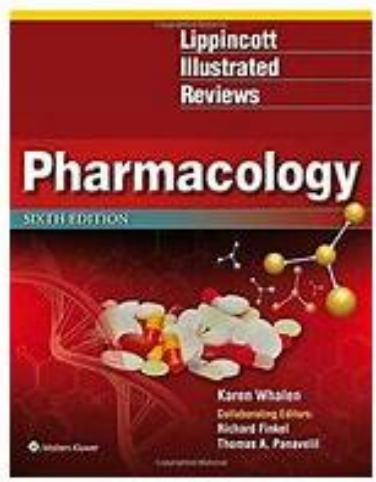
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